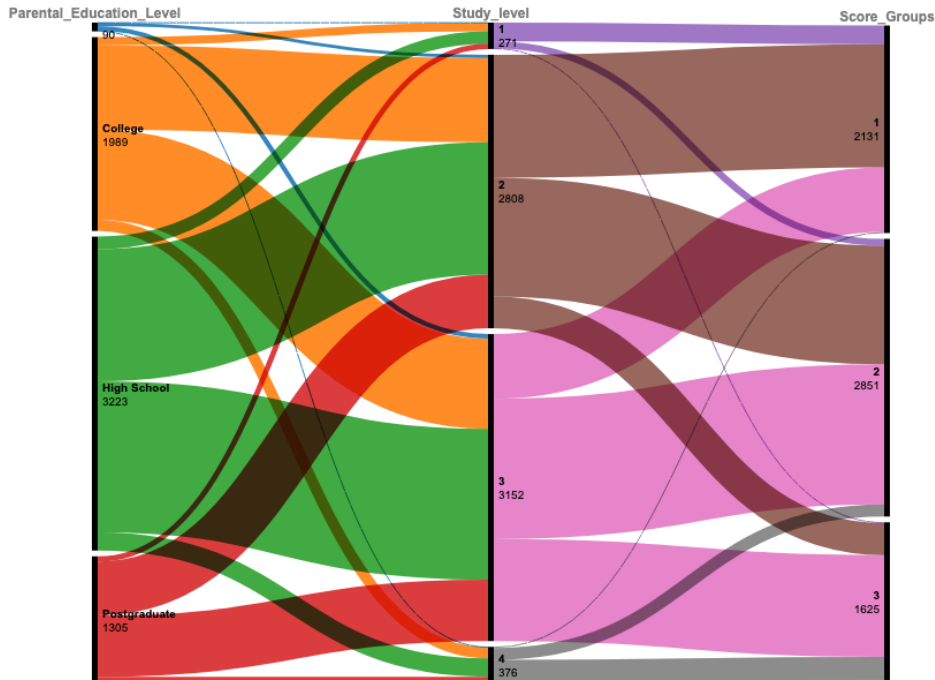


Data Visualization Pitch

Student Individual Assignment



The impact of different factors on student performances

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Research questions

Primary research question:

To what extent do study hours, perceived teacher quality, and family background influence students' exam performance?

Secondary research question:

- Can students' performance be explained solely by the number of hours they spend studying?
- Is there an association between perceived teacher quality and student performance?
- Can we find a pattern in students' background that shows which students tend to perform better?

About Data

This dataset contains information on 6,607 students and includes 20 attributes related to their personal habits, previous academic performance, and family background.

LICENSE:

Data from <https://www.kaggle.com/datasets/ayeshaseherr/student-performance> with CC BY-SA 4.0 license:

- -BY: you can share and adapt the work as long as you give appropriate credit to the author.
- -SA: you can share and adapt the work as long as you share the work you derive from it with the same license.

DATA QUALITY AND CRITICALITY:

The cleaning of this dataset did not require any action since there weren't any NA but while studying the variable "Teacher Quality" there were 78 empty records; while representing this variable, those records were excluded.

The main criticality encountered during the acquisition of the data is that The Student Performance Factors dataset, obtained from Kaggle, does not specify an original data collection methodology or specific population. This means that we can still understand some interesting insights about students' performance but we cannot locate them within any specific geographical area.

Methodology

This dataset was downloaded manually from Kaggle while the processing and cleaning phase was carried out with RStudio using the libraries Tidyverse and dplyr while the plots were made with Florish, Tableau and RAWgraphs.

The entire analysis is documented through code, ensuring full reproducibility. The R script and preprocessed data are available at <https://github.com/RF-unimib/Data-Visualization-Project.git>

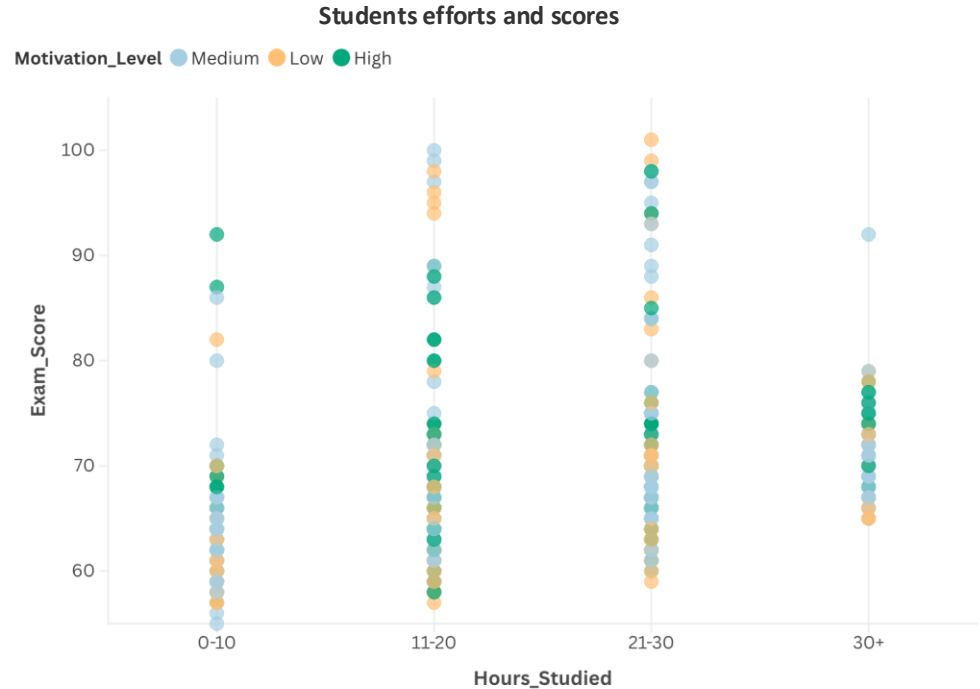
Data Transformation: To have a more meaningful representation, three numerical variables (Hours_studied, Previous_scores, Exam_score) were transformed by discretizing their continuous values into categorical ranges. This operation was performed to group observations into meaningful intervals, facilitating the visualization of patterns within specific value ranges.

- **Hours_studied** varies from 1 to 44 and was divided into four groups, from 1-10, 11-20, 21-30, 31-44 and each group was respectively associated the numbers 1 through 4.
- **Previous_scores** varies from 50 to 100 and to create three different classes we analyzed the quantiles to determine which thresholds to use. With the first quantile being 63 and third quantile being 88, the students with a grade in between the quantiles were assigned the value 2, the ones below the first 1 and the ones with a grade higher than 88, 3.
- **Exam_score** varies from 55 to 101 and, as it's a different exam, it's necessary to re-evaluate and change the thresholds of the different groups. Compared to the previous scores these grades were significantly lower; in fact the third quantile drops down to 69 while the first slightly changes to 65. The values assigned are now 1 to grades lower than 65, 2 to grades between 65 and 69 and 3 for the remaining values.

Positive impact of study hours on students' performance

This scatter plot shows exam scores grouped by how many hours they studied and colored by the student motivational level.

- We can see that usually, as expected, the more hours students spend studying, the higher the score tends to be.
- However, the variable Motivation Level does not show a clear pattern, we expect to see more green as we move to the right but it's equally distributed in every column.
- This visualization certainly suggests that there is a positive correlation between hours and score but it also suggests that other factors may influence students' results.



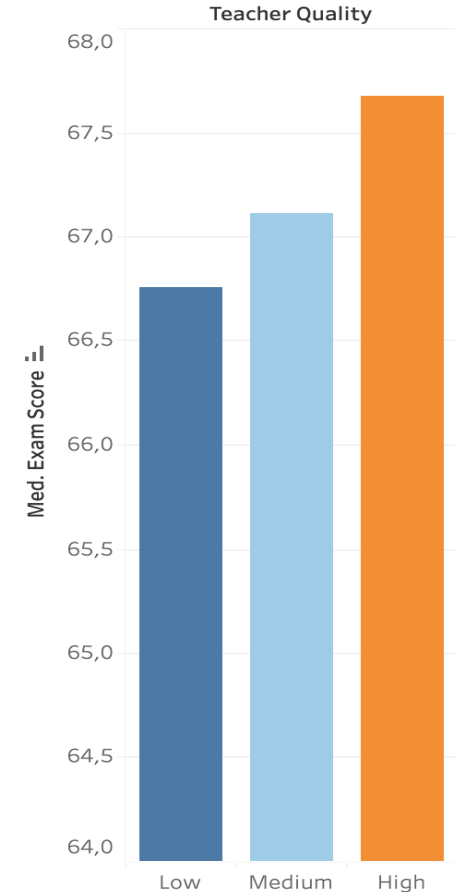
Perceived Teacher Quality effects positively Exam Scores

The chart shows a clear increasing trend in the average exam score across the three levels of perceived teacher quality.

- Students who report a high level of teacher quality achieve the highest average score, followed by those reporting medium quality, while the lowest average performance is observed among students perceiving low teacher quality.
- This pattern suggests a positive association between perceived teaching quality and student performance: higher perceived quality of instruction is associated with higher average exam scores.
- Even if the distance between the different averages is not very large, given that it is calculated using more than 6,000 records this result highlights that teachers still play a key role in having good results.
- In light of what was discussed earlier if the teacher leaves a good impression of themselves and of the subject to the students, they will be encouraged to spend more time studying, which eventually tends to lead to better results.

Bar chart made with Tableau

Teacher quality perceptions and grades

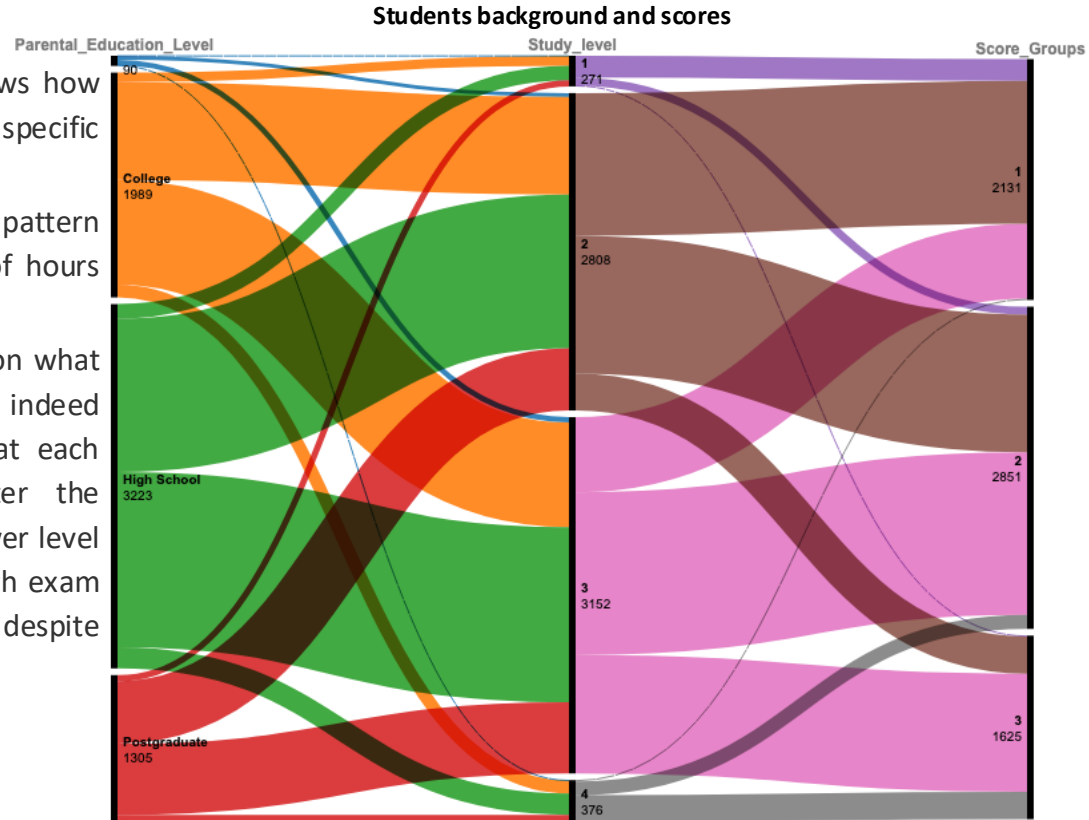


Students path towards great results

This alluvial diagram has 3 different variables and shows how many records for each value of a variable transition to a specific value in the subsequent variable.

This graph appears cluttered and does not reveal a clear pattern between the parents' educational level, the amount of hours studied by their children and their performances.

Given this fact it often happens that in data visualization what isn't shown in the graph is just as important as what is; indeed the fact that there isn't any real pattern means that each student is different from every other, no matter the background. In fact, students whose parents have a lower level of education may still study extensively and achieve high exam scores, while other students may obtain poor results despite having spent many hours studying.



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